SOFTWARE REQUIREMENTS DOCUMENT (SRD)

NASA Aqua–MODIS Satellite Data Acquisition & Processing Module

1. Purpose

The purpose of this module is to acquire and process satellite observations from the **MODIS (Moderate Resolution Imaging Spectroradiometer)** instrument onboard **NASA's Aqua satellite**. The processed satellite information will support the **AI-Enabled Antarctic Sea Ice and Iceberg Trajectory & Navigation Decision Support System**.

MODIS data will be used to obtain information related to **sea ice, iceberg regions, ocean conditions, surface characteristics, cloud cover, and temperature**. This information will be processed and supplied to the AI/ML and navigation decision-support modules.

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2. Satellite and Instrument

Component	Specification
Satellite	NASA Aqua
Instrument	MODIS
Full Form	Moderate Resolution Imaging Spectroradiometer
Data Type	Satellite imagery and scientific datasets
Coverage	Global
Primary Role	Earth observation and environmental monitoring
System Role	Remote-sensing data source

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3. Tools and Technologies

Tool / Technology	Function in the System
NASA Earthdata	Search, access and obtain NASA satellite datasets
NASA LAADS DAAC	Access and download MODIS products
MODIS Level-1B	Provides calibrated and geolocated radiance observations
MODIS Level-2 Products	Provides derived environmental/geophysical parameters
MODIS Level-3 Products	Provides spatially/temporally aggregated observations
Python	Main language for data acquisition, processing and AI/ML integration
GDAL	Raster/geospatial data processing and conversion
Rasterio	Reading, cropping, reprojecting and processing raster data
NumPy	Numerical and pixel-array operations
OpenCV	Image enhancement, segmentation and computer vision
AI/ML Frameworks	Iceberg/sea-ice detection and trajectory prediction
GPS	Provides real-time ship position
ECDIS	Displays navigation information and decision-support outputs

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4. NASA Earthdata

Function

NASA Earthdata provides access to NASA Earth-observation datasets, including MODIS Aqua observations.

System Requirements

The system shall use NASA Earthdata services to:

- Search for required MODIS datasets.

- Select NASA Aqua observations.
- Specify Antarctic geographic regions.
- Select required dates and observation periods.
- Retrieve required satellite data.
- Obtain satellite metadata.

Input

- Geographic coordinates/bounding box
- Date and time
- MODIS product
- Satellite/instrument

Output

- MODIS datasets
- Satellite imagery
- Observation metadata

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5. NASA LAADS DAAC

Function

LAADS DAAC provides access to MODIS data products and supports product discovery and data retrieval.

System Usage

It will be used for:

- MODIS product selection
- Data searching
- Data downloading
- Time-based data retrieval
- Accessing MODIS observations

The acquired data will be transferred to the local preprocessing module for further analysis.

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6. MODIS Data Products

Level-1B

Provides calibrated and geolocated radiance data.

Use:
Satellite image generation, spectral analysis and initial feature extraction.

Level-2

Provides derived geophysical/environmental parameters.

Potential Use:
Sea-surface temperature, cloud information and other environmental parameters.

Level-3

Provides aggregated spatial/temporal observations.

Use:
Regional analysis, historical analysis and AI/ML training datasets.

The final products shall be selected according to the required **spatial resolution, temporal resolution, Antarctic coverage and AI/ML requirements**.

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7. MODIS Data Processing

Python will be used as the primary programming language for the MODIS processing pipeline.

GDAL

GDAL will provide:

- Raster data reading
- Format conversion
- Geographic transformation
- Spatial cropping
- Resampling
- Geo-referencing
- Metadata extraction

Rasterio

Rasterio will be used for:

- Reading satellite raster data
- Reading individual bands
- Cropping the Antarctic region
- Reprojection
- Raster metadata handling

NumPy

NumPy will be used for:

- Pixel-level calculations
- Array manipulation
- Numerical operations
- Normalization
- Feature calculations

OpenCV

OpenCV may be used for:

- Image enhancement
- Noise reduction
- Thresholding
- Segmentation
- Edge/contour detection
- Computer-vision-based ice detection

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8. MODIS Processing Workflow

NASA Aqua
↓
MODIS
↓
NASA Earthdata / LAADS DAAC
↓
Data Search & Retrieval
↓
Data Validation
↓
GDAL / Rasterio
↓
Spatial & Raster Processing
↓

NumPy / OpenCV
↓
Image Enhancement & Feature Extraction
↓
Sea-Ice / Iceberg Detection
↓
AI/ML Model
↓
Iceberg Trajectory Prediction
↓
Risk Assessment
↓
Navigation Decision Support
↓
ECDIS

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9. Data Features

The system shall extract relevant features from MODIS observations.

Spatial Features

- Latitude
- Longitude
- Iceberg position
- Ice-covered area
- Iceberg boundary
- Distance from ship

Spectral/Image Features

- Reflectance
- Brightness
- Spectral information
- Surface characteristics

Environmental Features

- Sea-surface temperature
- Cloud information
- Surface conditions

Temporal Features

- Observation timestamp
- Previous iceberg position
- Current iceberg position
- Historical movement

These features will be provided to the AI/ML module for detection and trajectory prediction.

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10. Online and Offline Operation

Online Mode

When satellite/internet connectivity is available:

NASA Data Service
↓
MODIS Aqua Data
↓
Ship System
↓
Processing
↓
AI/ML Prediction
↓
Navigation Support

Offline Mode

When connectivity is unavailable:

Previously Downloaded MODIS Data
↓
Local Storage
↓
Local Processing
↓
AI/ML Prediction
↓
Navigation Decision Support

The system shall synchronize newly available data when connectivity is restored.

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11. Storage Requirements

To minimize storage requirements, the system shall avoid permanently storing unnecessary raw satellite imagery.

The recommended approach is:

Raw MODIS Data
↓
Temporary Processing
↓
Required Region/Product Extraction
↓
Processed Data
↓
Feature Dataset
↓
AI/ML Model

The system should retain:

- Required satellite metadata
- Processed observations
- Extracted features
- Iceberg coordinates
- Observation timestamps
- Prediction results

Large raw datasets should be retained only when required for future processing or model training.

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12. Functional Requirements

ID	Requirement
MOD-FR-01	System shall access MODIS Aqua datasets.

MOD-FR-02	System shall support Antarctic region selection.
MOD-FR-03	System shall support date/time-based data retrieval.
MOD-FR-04	System shall validate downloaded satellite data.
MOD-FR-05	System shall preprocess MODIS data.
MOD-FR-06	System shall perform required geospatial operations.
MOD-FR-07	System shall extract relevant satellite features.
MOD-FR-08	System shall provide processed data to the AI/ML module.
MOD-FR-09	System shall support offline processing of stored data.
MOD-FR-10	System shall integrate processed information with the navigation module.

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13. Non-Functional Requirements

Performance: Satellite data should be processed efficiently by limiting processing to the required Antarctic region and products.

Reliability: Satellite datasets shall be validated before being used by AI/ML models.

Storage Efficiency: Only required products, metadata, processed data and features should be retained locally.

Offline Capability: Previously acquired satellite data shall remain available when satellite connectivity is unavailable.

Scalability: The architecture should allow additional satellite data sources to be integrated in the future.

Maintainability: MODIS acquisition, processing and AI/ML components should remain modular.

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14. Role of MODIS in the Overall System

MODIS is the **satellite observation/data-source layer** of the system. It does not directly perform navigation decisions.

MODIS Aqua
↓
Satellite Observation
↓
Sea-Ice / Iceberg Information
↓
Feature Extraction
↓
AI/ML Detection & Prediction
↓
Iceberg Trajectory
↓
Risk Assessment
↓
Navigation Recommendation
↓
GPS + ECDIS

Final Technology Stack

NASA Aqua + MODIS → NASA Earthdata/LAADS DAAC → Python → GDAL/Rasterio → NumPy/OpenCV → AI/ML → GPS → ECDIS

This module therefore provides the **satellite-based environmental intelligence** required by the Antarctic iceberg trajectory and navigation decision-support system.